CityPersons Pedestrian Augmentation
using Stable Diffusion 3.5 Medium

Technical Report

Pipeline	Context-Person Composite (V3)
Backbone	Stable Diffusion 3.5 Medium
Segmenter	YOLOv8m-seg
Hardware	Kaggle T4 ×2, 448×448 resolution
Dataset	CityPersons (YOLO format, train split)

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1 Overview

The V3 Context-Person Composite pipeline generates photorealistic pedestrian augmentations for street scenes by decoupling generation from background preservation. SD3.5 generates candidate pedestrians; YOLOv8m-seg extracts the person mask; only person pixels are composited onto the original frame. This architectural separation ensures background integrity regardless of the model's output.

2 V3 Architecture

2.1 Core Pipeline

For each sample, V3 executes the following stages:

1. **Smart Placement:** Rule-based bounding-box selection respects road Y-range (0.66–0.92), perspective scale, a 5×5 slot grid with jitter, and avoids overlaps with existing annotations.
2. **Insertion Guide & Background Neutralisation:** A faint human silhouette ($\alpha = 0.44$) is painted at the target. The region is partially blended to neutral grey (strength 0.55), and context is darkened 3% to maintain texture cues.
3. **SD3.5 Img2Img:** Full-resolution frame (448×448) with T5 disabled for VRAM efficiency; model CPU offload and VAE/attention slicing enabled.
4. **Person Segmentation:** YOLOv8m-seg runs at confidence threshold 0.12; the detection closest to target with highest composite score is selected. Detection bbox is expanded 2.20× for spatial search.
5. **Mask Validation:** Checks for completeness (height coverage, aspect ratio 1.35–5.2), border contact ($>$ 6 px from edge), and minimum area (0.00045 of image).
6. **Ghost Detection:** Pixel contrast between source and generated image under mask must exceed 12.0 (0–255 scale). Three checks: early (under mask), mask area, and final (post-paste).
7. **Adaptive Retry:** Up to 3 attempts. Rejection reasons trigger specific parameter and prompt adjustments.
8. **Appearance Harmonisation:** Six-stage pipeline: local colour transfer (0.72), brightness matching (0.72), contrast matching (0.38, ratio [0.78, 1.16]), saturation matching (ratio [0.72, 1.12]), sensor noise injection (if $>$ 8% cleaner), and sharpness matching (sigma 0.60). Additional colour-temperature pass and edge-halo neutralisation (Gaussian estimate, radius 7.0).
9. **Contact Shadow:** Perspective-scaled elliptic shadow blended below feet; opacity scales from 16 (distant) to 46 (near).
10. **Output & Logging:** Final composite, side-by-side PNG, five-panel debug strip

(first 24 samples), and manifest CSV with all generation parameters and metadata.

3 Generation Parameters

Table 1: Per-variant hyperparameters.

Variant	Strength	Guidance	Steps	Weight
Single pedestrian	0.72	6.8	36	0.24
Two pedestrians	0.74	6.9	36	0.18
Small group (3)	0.76	7.0	38	0.16
Occluded pedestrian	0.74	6.8	36	0.18
Distant pedestrian	0.68	6.6	34	0.14
Near pedestrian	0.76	7.3	38	0.10

Strength controls deviation from input. Multi-person variants require higher strength (more new content generation). Guidance scale enforces prompt adherence; near-pedestrian uses highest (7.3) because scale and foot contact are hardest to achieve in foreground. Steps balance quality (higher) with runtime (lower); default 36 across variants.

4 Adaptive Retry Strategy

Table 2: Parameter and prompt adjustments per reject reason.

Reason	Parameter change	Prompt adjustment
Ghost / low contrast	Strength +0.04, Guidance +0.60 (cap 8.2)	Add "solid visible person, clear silhouette"; remove "transparent, faded"
Not enough persons	Strength +0.03, Guidance +0.55 (cap 8.6)	Add "requested count visible, separated silhouettes"; remove "single person, merged"
Too large	Strength −0.04, Guidance −0.60 (floor 6.0)	Add "mid-distance pedestrian"; remove "foreground close-up"
Cropped body	Strength −0.03, expand context 14%	Add "entire body visible, space around body"; remove "cropped"

Up to 3 retries per sample. Each retry adapts both generation strength/guidance and prompt terms based on the specific rejection reason. Samples exhausting all retries are discarded (no fallback).

5 Quality Filters

5.1 Scale Enforcement

Two gates:

- Early gate (YOLO detection on generated image): height ratio [0.68, 1.32]
- Final gate (post-paste): height ratio [0.76, 1.30]

Per-variant minimum thresholds: single 0.075, near 0.160, distant 0.065.

5.2 Ghost-Person Detection

Three checks prevent low-contrast insertions:

- Mean pixel difference under mask > 12.0 (0–255 scale)
- Segmented mask area > 0.00045 of image

- Final contrast check on composite > 0.010

5.3 Multi-Person Count Validation

For variants requiring multiple persons, pipeline explicitly verifies expected count was generated. If fewer detected, sample is rejected with reason "not_enough_new_people", triggering high-strength adaptive retry.

5.4 Mask Quality

Validated for:

- Vertical band coverage > 0.24 (head, torso, legs all present)
- Aspect ratio [1.35, 5.2] (consistent with standing/walking person)
- Border contact: masks within 6 px of image edge rejected
- Spatial containment: mask outside detection bbox < 0.22

6 Smoke Test Results

6.1 Configuration

10 images (multi-person variants only: two-pedestrian and small-group).

6.2 Outcome

4 accepted outputs out of 10 attempted (40% acceptance rate).

6.3 Reject Reasons

- **low_person_conf** (dominant): After scale failure on attempt 1, adaptive retry reduced strength and guidance, causing further confidence drop on subsequent attempts. **Linked failure chain:** scale reduction cascades to confidence failure below 0.35 threshold.
- **too_large_for_perspective**: SD3.5 generated persons with height ratio exceeding max (1.32). Adaptive retry correctly reduced strength but lower-energy generation fell below confidence threshold.
- **floating_or_bad_ground**: One rejection for inadequate foot-ground contact (lack of contact shadow or improper positioning).

6.4 Accepted Samples

- bremen_000209 (two-ped)
- aachen_000123 (two-ped)
- bremen_000109 (two-ped)
- bremen_000186 (small-group)

Multi-person small-group had lower acceptance rate than two-pedestrian, consistent with increased difficulty of generating three coherent persons at correct scale.

7 Technical Findings

1. **Architectural separation of generation and preservation.** Background preservation cannot be enforced via prompting alone. The only reliable solution is architectural: take background from the original frame; composite only the generated person pixels. This guarantee holds regardless of SD3.5's output.
2. **Scale freedom with faint guide outperforms strict constraint.** A loose scale envelope (30% wider, 16% taller than target) with faint insertion guide ($\alpha = 0.44$) achieves better person quality while still biasing the model toward correct size/location. Strict bounding-box constraints produce ghost silhouettes; loose envelopes permit the model generative freedom.
3. **Ghost detection via pixel contrast is non-negotiable.** Without pixel contrast validation, a non-trivial fraction of accepted samples contain near-transparent "ghost" persons invisible to downstream detectors. The 12.0 (0–255) threshold is empirically validated.
4. **Adaptive retry with reason-specific adjustments outperforms fixed escalation.** A fixed strength increase applied to an over-scaled person makes it even larger on the next attempt. Reason-aware retry (decrease strength for over-scale, expand context for cropped bodies, targeted prompts for ghosts) recovers from failures that fixed escalation cascades on. The smoke test reveals a linked failure chain: scale-retry reduction → confidence failure. Future refinement: widen confidence threshold on retry 2 after scale failure rather than applying full guidance reduction.
5. **Multi-person count must be explicitly validated.** A two-pedestrian variant generating one person represents a scenario failure, not a fallback. Explicit count validation prevents silent acceptance of wrong configurations.

8 Next Steps

8.1 Production Run

1. Run 200-image batch across all six variants on train split.
2. Inspect debug strips for patterns in ghost, scale, and multi-person count rejections.
3. If `low_person_conf` > 30% after scale-retry: widen confidence threshold on retry 2 following `too_large_for_perspective` (accept 0.28 instead of 0.35).
4. If multi-person count failure > 40%: review insertion guide geometry for small-group case.

5. Enable post-hoc metrics (SSIM, PSNR, histogram distance, brightness/contrast delta per variant).

8.2 Quality Gate

- Target: acceptance $\geq 80\%$ and background SSIM ≥ 0.86 across all variants.
- If acceptance $< 60\%$ after parameter sweep: LoRA fine-tuning on CityPersons pedestrian crops.

9 Summary

V3 resolves background preservation through architectural separation: SD3.5 generates freely, segmentation extracts the person, original background is always used in the final composite. Quality is ensured by scale gates, ghost detection, multi-person count validation, and appearance harmonisation. The smoke test identified a linked failure chain (scale-retry $\rightarrow$ confidence failure) pointing to specific parameter refinements. A 200-image production run is required to validate against the 80% / 0.86 SSIM targets.

References

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